

CleverCrest AI

Product Development Roadmap & Feature Requirements

Version-wise • Phase-wise • MVP to Future Extensions

Project Type: Multi-Tenant Enterprise Knowledge Intelligence Platform

Primary Stack: React + TypeScript • FastAPI + Python • PostgreSQL • ChromaDB • OpenRouter

Development Strategy: MVP-first, modular, low-cost/open-source, extension-ready

Development Principle

Build a small, reliable Version 1.0 first. Every core module must have clean boundaries so future capabilities can be added as independent extensions without destabilizing the MVP.

1. Document Purpose

This document defines what will be developed, in what order, and what is intentionally deferred.

CleverCrest AI will be developed as a modular product rather than a one-off chatbot. The roadmap separates product maturity (Version) from implementation order (Phase). Version 1.0 is the semester MVP and contains only the capabilities required to demonstrate a complete, secure, end-to-end knowledge-to-AI workflow.

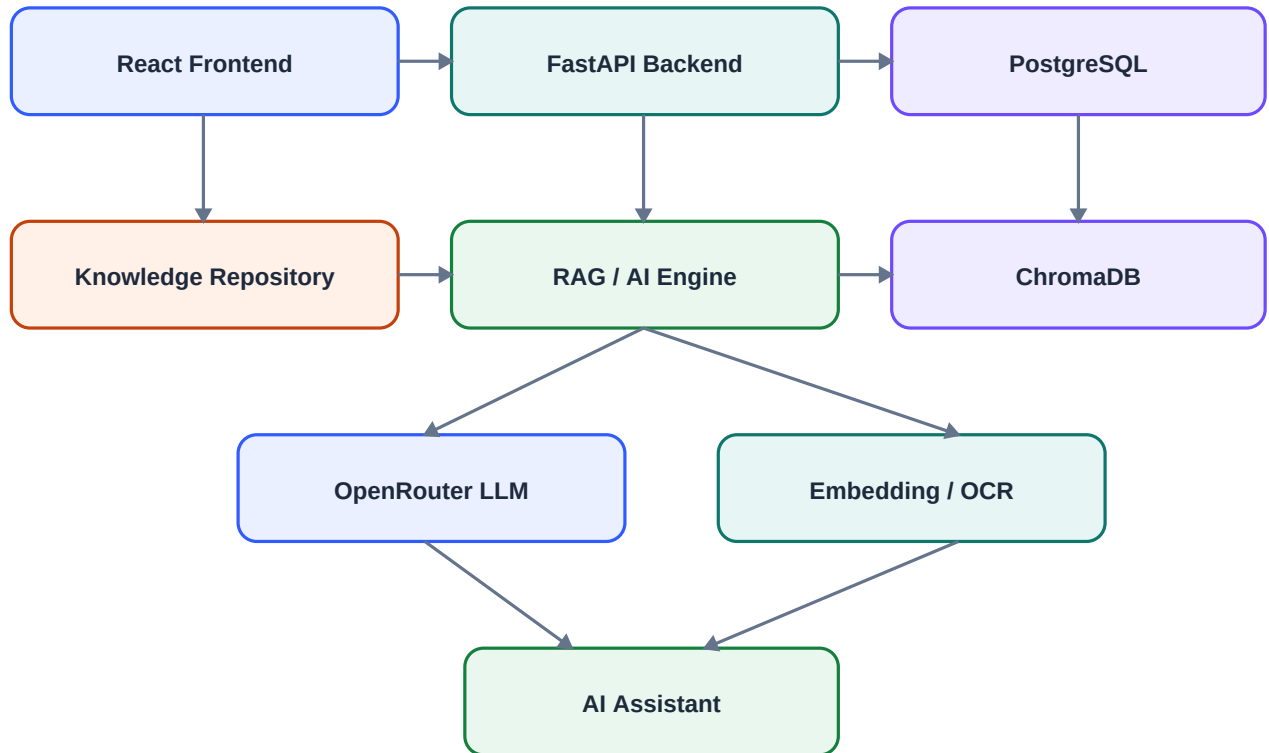
Item	Definition
Version	A product maturity level such as MVP, Enterprise Enhancement, AI Intelligence, or SaaS Product.
Phase	A development block inside a version, implemented in dependency order.
MVP	The smallest complete version that is useful, demonstrable, secure, and deployable.
Future Extension	A planned capability designed to plug into the core platform later without forcing a rewrite.

2. Development Principles

- **MVP first:** finish the core workflow before adding advanced AI features.
- **Accuracy over feature count:** each released module must work correctly and be testable.
- **Modular architecture:** authentication, RBAC, knowledge, RAG, AI, notifications, and analytics should have clear service boundaries.
- **Secure by default:** tenant isolation and permission checks must occur on the server, not only in the UI.
- **Approved knowledge only:** a document must pass the configured governance workflow before it can influence AI responses.
- **Low-cost implementation:** prefer open-source models, local embeddings, free tiers, and replaceable providers.
- **Extension-ready:** future features should consume stable services and APIs instead of changing the foundation unnecessarily.

3. Overall Product Architecture

The product is organized around a secure knowledge lifecycle. Users interact primarily with the AI Assistant, while administrators and management roles operate the knowledge governance side of the platform.



Core rule: the AI Assistant is a capability of the Knowledge Intelligence Platform, not the entire product. End users do not receive direct access to confidential source files.

4. Knowledge-to-AI Lifecycle



Approved knowledge → semantic retrieval → grounded AI response

- **Upload:** an authorized contributor submits a document and metadata.
- **Governance:** the document remains outside the AI knowledge base until approved.
- **Processing:** text is extracted, OCR is applied where required, and content is cleaned.
- **Chunking:** the content is divided into retrieval-friendly sections while retaining document metadata.
- **Embedding:** each chunk is converted into a numerical vector using an open-source embedding model.
- **Indexing:** vectors and metadata are stored in ChromaDB.
- **Retrieval:** a user question is converted to an embedding and matched against only authorized knowledge.
- **Generation:** retrieved context is passed to an LLM through OpenRouter to generate a grounded response.

5. VERSION 1.0 — MVP

Version 1.0 Goal

Deliver a complete, deployable platform covering authentication, multi-tenancy, dynamic RBAC, secure knowledge management, approval workflow, document processing, vector retrieval, AI assistance, permission requests, notifications, audit logs, and basic dashboard/search utilities.



Phase 1 — Platform Foundation

Establish the application infrastructure and identity model that every other module depends on.

1.1. Authentication

Purpose: Provide secure user identity and session management.

Version 1.0 Requirements

- Registration and login
- JWT access authentication
- Password hashing
- Logout
- Forgot/reset password
- Email verification
- Basic session handling

Future Extension: Refresh tokens, Google/Microsoft OAuth, SSO, MFA.

1.2. Organization Management

Purpose: Create the tenant boundary for each company, institute, or organization.

Version 1.0 Requirements

- Create/update organization profile
- Organization logo and basic details
- Organization status
- Organization settings/preferences
- Tenant-aware backend context

Future Extension: Branches, subscriptions, billing, tenant-level feature flags.

1.3. Department Management

Purpose: Organize users and knowledge by functional area.

Version 1.0 Requirements

- Department CRUD
- Department description
- Department head/manager assignment
- Department membership

Future Extension: Nested departments and advanced organizational hierarchy.

1.4. User Management

Purpose: Allow authorized administrators to control membership.

Version 1.0 Requirements

- Invite user
- Activate/deactivate user
- Suspend user

- Delete user
- Profile management
- User status/search/filter

Future Extension: Bulk import, directory synchronization, LDAP/Azure AD.

Phase 2 — Enterprise RBAC

Implement dynamic roles and permissions so organizations can model their own working structure.

2.1. Role Management

Purpose: Allow organization admins to create custom management roles.

Version 1.0 Requirements

- Create/update/delete roles
- Assign role to users
- Role descriptions
- Support roles such as Project Manager, Reviewer, Faculty, Trainer, HR

Future Extension: Role templates and inherited roles.

2.2. Permission Engine

Purpose: Enforce permissions consistently across backend APIs and UI.

Version 1.0 Requirements

- Upload documents
- Review/approve/reject
- Delete documents
- Create/manage collections
- Manage users/departments
- View audit logs
- Manage notifications/AI settings

Future Extension: Attribute-Based Access Control and more granular resource policies.

2.3. Access Requests

Purpose: Allow users to request access when an AI query requires knowledge outside their current permissions.

Version 1.0 Requirements

- Create access request
- Record requested collection/resource
- Admin/reviewer approve or reject
- Store decision and audit trail
- Re-evaluate permissions after approval

Future Extension: Temporary access, automatic expiry, delegated approval.

Phase 3 — Knowledge Repository

Build the central internal repository from which approved AI knowledge will be created.

3.1. Collections

Purpose: Group documents into meaningful internal knowledge domains.

Version 1.0 Requirements

- Collection CRUD
- Department association
- Owner
- Description
- Tags
- Status/visibility rules

Future Extension: Nested collections and richer taxonomy.

3.2. Document Management

Purpose: Store and manage source files plus metadata without exposing them to ordinary end users.

Version 1.0 Requirements

- PDF, DOCX, PPTX, TXT, image, and Excel intake
- Upload/replace/archive/delete
- Owner
- Department
- Collection
- Tags
- Language
- Version
- Expiry/status metadata

Future Extension: Audio/video ingestion, ZIP processing, richer metadata extraction.

3.3. Processing Queue

Purpose: Move document processing into explicit states so uploads do not block the user interface.

Version 1.0 Requirements

- Pending
- Processing
- OCR
- Chunking
- Embedding
- Completed
- Failed
- Retry

Future Extension: Priority queues and distributed background workers.

Phase 4 — Knowledge Governance

Control which information is trusted enough to become AI knowledge.

4.1. Approval Workflow

Purpose: Ensure only reviewed and approved material can be indexed.

Version 1.0 Requirements

- Draft
- Submitted
- Under Review
- Rejected
- Approved
- Published
- Indexed
- Approve/reject comments
- Revision request
- Approval timeline

Future Extension: Multiple reviewers, multi-level approval, delegated approval.

4.2. Audit Logs

Purpose: Create a traceable record of security and knowledge-management actions.

Version 1.0 Requirements

- Login events
- Upload/delete
- Approve/reject
- Role changes
- Permission changes
- Collection changes
- Important AI configuration changes

Future Extension: Exportable compliance reports and retention rules.

Phase 5 — AI Knowledge Engine

Convert approved documents into a permission-aware retrieval system.

5.1. OCR Engine

Purpose: Extract machine-readable text from scanned PDFs and images.

Version 1.0 Requirements

- Detect whether OCR is required
- Extract text
- Preserve basic page/document metadata
- Handle OCR failure

Future Extension: Handwriting OCR and specialized document recognition.

5.2. Document Processing

Purpose: Normalize extracted content before indexing.

Version 1.0 Requirements

- Cleaning
- Whitespace/format normalization
- Chunking
- Metadata preservation
- Document-level identifiers

Future Extension: Semantic/smart chunking and structure-aware parsing.

5.3. Embedding Engine

Purpose: Convert knowledge chunks and queries into vectors.

Version 1.0 Requirements

- Select open-source embedding model
- Generate embeddings
- Attach metadata
- Update embeddings when content changes
- Delete embeddings when knowledge is removed

Future Extension: Multiple embedding models and per-tenant model selection.

5.4. Vector Database

Purpose: Store and retrieve embeddings efficiently.

Version 1.0 Requirements

- ChromaDB collections/indexes
- Vector insert/update/delete
- Top-K similarity search
- Metadata filtering

Future Extension: Qdrant or another scalable vector store.

5.5. Semantic Search

Purpose: Retrieve relevant knowledge using meaning rather than exact keywords.

Version 1.0 Requirements

- Query embedding
- Top-K retrieval
- Collection filtering
- Tenant filtering
- Permission filtering
- Basic ranking

Future Extension: Hybrid keyword + vector search and reranking.

5.6. Prompt Builder

Purpose: Construct a grounded LLM request from authorized context.

Version 1.0 Requirements

- User question
- Conversation context
- Retrieved chunks
- Organization/system instructions
- Response constraints

Future Extension: Reusable prompt templates and configurable organization policies.

5.7. AI Response Engine

Purpose: Generate grounded answers through an interchangeable LLM provider.

Version 1.0 Requirements

- OpenRouter integration
- Free/low-cost model support
- Response generation
- Streaming where practical
- Source metadata
- Basic confidence/quality indicator

Future Extension: Multiple model routing, fallback models, provider abstraction.

Phase 6 — AI Assistant

Provide the end-user interaction layer over the knowledge engine.

6.1. Chat

Purpose: Provide a simple, focused AI Assistant interface.

Version 1.0 Requirements

- New conversation
- Send questions
- Receive answers
- Follow-up questions
- Rename chat
- Delete chat
- Chat history

Future Extension: Pinned chats, folders, saved conversations.

6.2. Conversation Memory

Purpose: Maintain enough recent context for natural follow-up questions.

Version 1.0 Requirements

- Session context
- Previous turns
- Conversation identifiers
- Controlled context window

Future Extension: Long-term memory and user preference memory.

6.3. Permission-Aware AI

Purpose: Prevent unauthorized knowledge retrieval before context reaches the LLM.

Version 1.0 Requirements

- Check tenant
- Check user permissions
- Filter collections/documents
- Offer access request when needed
- Record decision

Future Extension: Chunk-level policies and advanced policy engines.

6.4. Response Metadata

Purpose: Give users limited transparency without exposing confidential documents.

Version 1.0 Requirements

- Indicate that the response is based on approved organizational knowledge
- Show non-sensitive source title/version metadata where permitted
- Show basic confidence indicator

Future Extension: Richer explainability controls without revealing source content.

Phase 7 — Platform Utilities

Add the minimum operational tools required to make the MVP usable by administrators and management.

7.1. Notifications

Purpose: Inform users about important workflow events.

Version 1.0 Requirements

- Approval/rejection
- Access granted/rejected
- Processing completed
- System notifications
- Unread state

Future Extension: Real-time notifications and push delivery.

7.2. Dashboard

Purpose: Provide a basic operational overview.

Version 1.0 Requirements

- Document count
- Collection count
- User count
- Pending approvals
- Processing status
- Basic AI usage indicator

Future Extension: Interactive analytics and department-level insights.

7.3. Search

Purpose: Give management users direct tools to locate platform objects.

Version 1.0 Requirements

- Document search
- Collection search
- User search
- Role search
- Filtering

Future Extension: Global enterprise search. End users continue to rely primarily on the AI Assistant.

13. Version 1.0 — MVP Completion Criteria

Version 1.0 should be considered complete only when the complete knowledge lifecycle works end-to-end and the major security boundaries are enforced.

Area	Minimum completion condition
Tenant isolation	A user can access only the organization to which the account belongs.
RBAC	API endpoints reject actions for which the user lacks permission.
Knowledge governance	Unapproved documents cannot be retrieved by the AI.
Processing	An uploaded document moves through visible processing states and reaches indexed status or a clear failure state.
RAG	A permitted user question retrieves relevant approved context and generates a grounded response.
Access requests	A user without required knowledge access can submit a request and an authorized reviewer can decide it.
Auditability	Important security, role, knowledge, and approval actions create audit records.
Usability	Admin/management users can operate the knowledge workflow without developer intervention.
Deployment	The MVP can run using the selected low-cost/free infrastructure within practical resource limits.

14. VERSION 2.0 — Enterprise Enhancement

Version 2.0 improves operational intelligence and knowledge quality after the MVP is stable. These features should reuse the V1 APIs and services rather than changing the core tenant/RBAC/RAG architecture.

. Analytics & Insights

Purpose: Turn platform activity into actionable information.

Version 1.0 Requirements

- Most asked questions
- Knowledge gap detection
- Unused documents
- Popular topics
- Department statistics
- AI usage reports

Future Extension: Add an analytics service that consumes audit/chat/knowledge events; do not couple analytics calculations to the core chat request.

. Document Intelligence

Purpose: Improve the quality and maintainability of the knowledge repository.

Version 1.0 Requirements

- Duplicate detection
- Document similarity
- Automatic metadata extraction
- Smart tagging
- Language detection
- Version comparison

Future Extension: Add post-processing workers that consume the existing document-processing pipeline.

. Advanced Knowledge Management

Purpose: Support more realistic enterprise document lifecycles.

Version 1.0 Requirements

- Document versioning
- Scheduled publishing
- Scheduled expiry
- Knowledge lifecycle states
- Bulk upload
- Bulk approval

Future Extension: Extend document metadata and workflow state machines while preserving the same RAG indexing interface.

. User Experience Enhancements

Purpose: Improve productivity without changing core security.

Version 1.0 Requirements

- Bookmarks
- Saved answers
- Chat export
- User feedback
- Theme/dark mode
- Improved search dashboard

Future Extension: Implement these as frontend modules backed by stable APIs.

15. VERSION 3.0 — AI Intelligence

Version 3.0 introduces advanced AI capabilities only after the core RAG system, permissions, logging, and data model are proven stable.

. LangGraph Multi-Agent System

Purpose: Introduce specialized AI agents that can coordinate while still using the same permission-aware knowledge services.

Version 1.0 Requirements

- HR Agent
- Legal Agent
- Finance Agent
- Training Agent
- Research Agent
- Project Agent

Future Extension: Agents should call existing retrieval and policy services instead of directly bypassing authorization.

. Voice AI

Purpose: Allow natural voice interaction with the same knowledge engine.

Version 1.0 Requirements

- Speech input
- Speech-to-text
- Text-to-speech
- Voice assistant mode

Future Extension: Add a voice gateway that converts speech to the existing chat API rather than creating a second knowledge pipeline.

. Recommendation Engine

Purpose: Proactively suggest useful knowledge or next actions.

Version 1.0 Requirements

- Suggested collections
- Suggested questions
- Suggested follow-ups
- Knowledge recommendations

Future Extension: Build recommendations from approved metadata and usage events.

. Adaptive Learning

Purpose: Provide education-specific intelligence without changing the enterprise core.

Version 1.0 Requirements

- Student progress
- Weak-topic detection
- Personalized study plans

- Adaptive quizzes
- Revision planning
- Performance tracking

Future Extension: Implement as an Education extension consuming user progress data plus the existing RAG/AI services.

. Enterprise AI Automation

Purpose: Move from question answering toward assisted workflows.

Version 1.0 Requirements

- Meeting summaries
- Document summaries
- Policy change summaries
- Workflow assistance
- Email drafting

Future Extension: Expose automation as separate workflow services with explicit permission checks.

. Mobile Platform

Purpose: Provide mobile access while reusing backend APIs.

Version 1.0 Requirements

- React Native client
- Offline-safe cached UI where appropriate
- Push notifications
- Mobile chat

Future Extension: Build mobile as a new client of the existing APIs; do not duplicate backend business logic.

16. VERSION 4.0 — SaaS Product

Version 4.0 is a commercial-product direction. It is not required for the semester project. Its purpose is to show how CleverCrest could evolve into a real multi-tenant SaaS offering.

. Subscription & Billing

Purpose: Turn organizations into managed SaaS tenants.

Version 1.0 Requirements

- Subscription plans
- Usage quotas
- Payment gateway
- Trial management
- Organization onboarding

Future Extension: Add billing as an independent service so product features do not depend on payment logic.

. Enterprise Administration

Purpose: Provide platform-owner controls over all tenants.

Version 1.0 Requirements

- Super Admin dashboard
- Tenant lifecycle management
- Global analytics
- Feature flags
- Organization health monitoring

Future Extension: Add a separate platform-admin authorization layer above organization-admin permissions.

. Integrations

Purpose: Connect CleverCrest to systems where organizational knowledge already exists.

Version 1.0 Requirements

- Microsoft 365
- Google Workspace
- Slack
- Microsoft Teams
- SharePoint
- REST/Webhook integrations

Future Extension: Create connector adapters that normalize external content into the existing document-processing pipeline.

. Security & Compliance

Purpose: Raise the platform toward enterprise deployment requirements.

Version 1.0 Requirements

- SSO

- MFA
- Encryption/key management
- Data retention policies
- Compliance reports

Future Extension: Add security services around the existing identity, audit, storage, and tenant-isolation layers.

17. Extension Architecture

The most important architectural rule is that future modules must extend stable core services rather than rewrite them. Domain-specific features should consume common identity, organization, RBAC, collection, document, retrieval, AI, audit, and notification capabilities.

Layer	Examples	Extension Rule
Core Platform	Authentication, tenant isolation, RBAC, collections, repository, approval, RAG, AI Assistant	Must remain stable and centrally governed.
Education Extension	Student progress, quizzes, learning plans, adaptive tutoring	Uses core RAG + AI + user/profile services.
Corporate Extension	HR workflows, project knowledge, policy automation	Uses core collections, permissions, workflow and AI.
Healthcare Extension	Clinical SOP knowledge and controlled staff workflows	Uses strict permissions, audit, and approved knowledge.
Legal Extension	Contract/policy knowledge and review workflows	Uses document lifecycle, permissions and audit.
AI Extensions	Agents, voice, recommendations	Use existing AI gateway and retrieval services.

Golden rule: a new module should be able to call an existing service/API with minimal changes to unrelated modules. If adding a feature requires rewriting authentication, tenant isolation, or the RAG pipeline, the architecture should be reviewed before implementation.

18. Explicitly Out of Scope for Version 1.0

The following are intentionally deferred. They are not missing requirements; they are scope-control decisions designed to keep the semester MVP achievable and reliable.

- LangGraph multi-agent orchestration
- Voice assistant and speech processing
- Native mobile application
- Advanced recommendation engine
- Adaptive education engine
- Advanced analytics and knowledge-gap intelligence
- Handwriting OCR
- Video/audio knowledge ingestion
- Real-time collaboration
- SaaS billing and subscriptions
- Large-scale external integrations such as Microsoft 365, SharePoint, Slack, and Teams
- Enterprise SSO/MFA beyond the security required for the MVP

19. Recommended Implementation Order

The order below reflects technical dependencies. Do not start AI integration before tenant isolation, permissions, document lifecycle, and processing contracts are stable.

Order	Workstream	Primary Output
1	Project foundation	Repository, environment configuration, database connection, API structure, frontend shell
2	Authentication	Users, JWT, password/security basics
3	Multi-tenancy	Organization context and tenant isolation
4	RBAC	Roles, permissions, authorization middleware
5	Collections & documents	Knowledge repository and metadata
6	Approval workflow	Review states, approval actions, audit events
7	Processing pipeline	Extraction, OCR, cleaning, chunking, queue states
8	Vector layer	Embeddings, ChromaDB, metadata filters
9	RAG service	Retrieval, prompt construction, OpenRouter gateway
10	AI Assistant	Chat UI, history, memory, permission-aware responses
11	Utilities	Notifications, dashboard, management search
12	Hardening	Testing, security review, error handling, deployment, documentation

20. Quality Gates Before Moving Forward

- **Phase gate:** a phase should not be considered complete until its core user flow works end-to-end.
- **Security gate:** authorization must be enforced server-side for every protected resource.
- **Data gate:** tenant identifiers and ownership relationships must be present where isolation is required.
- **AI gate:** the RAG service must never retrieve knowledge that the requesting user is not allowed to use.
- **Failure gate:** failed document processing must be visible and retryable.
- **Audit gate:** important state-changing actions must be traceable.
- **Deployment gate:** environment variables, secrets, CORS, database migrations, storage, and model configuration must be reproducible.
- **Documentation gate:** APIs and major architectural decisions should be documented before the next major phase begins.

Completion Philosophy

The goal is not to maximize the number of modules. The goal is to complete a coherent, secure, accurate MVP that demonstrates a real enterprise knowledge lifecycle and can be extended cleanly when time or resources allow.

21. Final Product Roadmap Summary



Version	Focus	Status for Semester
1.0 — MVP	Core platform, RBAC, knowledge governance, RAG, AI Assistant, utilities	Primary build target
2.0 — Enterprise Enhancement	Analytics, document intelligence, advanced knowledge management, UX	Optional / after MVP
3.0 — AI Intelligence	Agents, voice, recommendations, adaptive learning, automation, mobile	Future roadmap
4.0 — SaaS Product	Billing, integrations, enterprise administration, security/compliance	Long-term vision

Project Name: CleverCrest AI

Core Positioning: Multi-Tenant Enterprise Knowledge Intelligence Platform with secure RAG-powered AI assistance.

Development Strategy: Build Version 1.0 completely, then extend through independent modules.

End of Product Development Roadmap & Feature Requirements